MidRC Part 4

Sample Problems

Exercise 1

Find the equation of the tangent line to the curve $x^2 + e^{xy} - y^2 = -3$ at the point $(x, y) = (0, 2)$.

Exercise 2

Evaluate the limits. Show your work.

(i)

$\lim_{t \to 2} \left(\frac{t^2-4}{t+1-\sqrt{4t+1}} \right)$

(ii)

$\lim_{t \to 0} \left[(\cos t)^{1/t^2} \right]$

Exercise 3

Consider $f(x) = \dfrac{e^{3x}(x^2+1)}{\sqrt[3]{x}\,\arctan(x)}$, and $\dfrac{\mathrm{d}f}{\mathrm{d}x} = f(x)g(x)$. Find $g(x)$ for $x > 0$ and evaluate $g(1)$.

Exercise 4

Given $f : \mathbb{R} \to \mathbb{R}, f(x) = \arctan(\sin(x))$, show that the mean value theorem works for f over the interval $[0, \frac{\pi}{2}]$. (You may assume that f is continuous and differentiable)

Exercise 5

Let $y = y(x)$ be implicitly defined by the equation $y^3 - 2y = 3x^2 - 4x$, decide the monotonicity (increasing/decreasing) and convexity (convex/concave) of the function $y = y(x)$ near the point $(x, y) = (1, 1)$.

Exercise 6

Given function $f : \mathbb{R} \to \mathbb{R}$ and

- $(f(0), f'(0)) = (6, -3/2)$
- $(f(5), f'(5)) = (3, -2/3)$
- $(f(8), f'(8)) = (0, -1/4)$
- $(f(1), f'(1)) = (5, -4/3)$
- $(f(7), f'(7)) = (1, -3/2)$

Calculate $g'(5)$ where $g = f^{-1} \circ f^{-1}$.

Exercise 7

Given $g : \mathbb{R} \to \mathbb{R}$

$$g(x) = \begin{cases} \frac{1}{3xe^{-2x}} - \frac{1}{x(x+3)}, & \text{if } x \neq 0 \\ a, & \text{if } x = 0 \end{cases}$$

where $a \in \mathbb{R}$. If g is continuous, find the value of a.

Exercise 8

Given function $f : \mathbb{R} \to \mathbb{R}$ with derivative $f'(x) = \left(\frac{x}{4} - 2\right)^3 (4 - x)$

1. Find all critical point(s) of f.
2. Determine where f is increasing or decreasing. (open intervals suffice)
3. Determine where f is convex or concave. (open intervals suffice)
4. Classify each critical point from (i) as a local maximum, a local minimum, or neither.
5. Find the inflection point(s) of f.

Exercise 9

Given a continuous function

$$f : [-2, 2] \to \mathbb{R} \quad f(x) = \frac{\sqrt[3]{x^2}}{x^2 + 2x + 3}$$

Find the global maximum and minimum over the domain of f. (Show your work. Put the final candidate points in the table below and indicate their features using ✓.)

x	$f(x)$	global min	global max	neither

Sample Solution

Exercise 1

Consider $y = y(x)$, apply $\frac{d}{dx}$ to both sides, we have

$$2x + e^{xy}\left(y + x\frac{dy}{dx}\right) - 2y\frac{dy}{dx} = 0$$

At $(x, y) = (0, 2)$, we have

$$2 \cdot 0 + e^{0\cdot 2}(2 + 0\frac{\mathrm{d}y}{\mathrm{d}x}) - 2 \cdot 2\frac{\mathrm{d}y}{\mathrm{d}x} \implies 2 + 0\frac{dy}{dx} - 4\frac{dy}{dx} = 0 \implies \frac{dy}{dx} = \frac{2}{4} = \frac{1}{2}$$

Therefore the tangent line is $y - 2 = \frac{1}{2}(x - 0)$, or $y = \frac{1}{2}x + 2$.
Remark: could also consider $x = x(y)$ & apply $\frac{dx}{dy}$.

Exercise 2

(i)

$$\lim_{t\to 2}(\dots) \overset{L'H}{=} \lim_{t\to 2}\frac{2t}{1 - \frac{2}{\sqrt{1+4t}}} = 12$$

Alternative method: direct calculation

$$\begin{aligned}
\lim_{t\to 2}(\dots) &= \lim_{t\to 2}\frac{(t+2)(t-2)(t+1+\sqrt{4t+1})}{(t+1-\sqrt{4t+1})(t+1+\sqrt{4t+1})} \\
&= \lim_{t\to 2}\frac{(t+2)(t-2)(t+1+\sqrt{4t+1})}{(t+1)^2 - (4t+1)} \\
&= \lim_{t\to 2}\frac{(t+2)(t-2)(t+1+\sqrt{4t+1})}{t^2+2t+1-4t-1} \\
&= \lim_{t\to 2}\frac{(t+2)(t-2)(t+1+\sqrt{4t+1})}{t^2-2t} \\
&= \lim_{t\to 2}\frac{(t+2)(t-2)(t+1+\sqrt{4t+1})}{t(t-2)} \\
&= \lim_{t\to 2}\frac{(t+2)(t+1+\sqrt{4t+1})}{t} \\
&= \frac{(2+2)(2+1+\sqrt{4\cdot 2+1})}{2} \\
&= \frac{4(3+\sqrt{9})}{2} \\
&= \frac{4(6)}{2} \\
&= 12
\end{aligned}$$

(ii)

Let $L = \lim_{t\to 0}\left[(\cos t)^{1/t^2}\right]$,
then $\log L = \lim_{t\to 0}\frac{\log(\cos t)}{t^2} \overset{L'H}{=} \lim_{t\to 0}\frac{-\sin t/\cos t}{2t} = -\frac{1}{2}\lim_{t\to 0}\frac{\sin t}{t} \cdot \lim_{t\to 0}\frac{1}{\cos t} = -\frac{1}{2} \cdot 1 \cdot 1 = -\frac{1}{2}$
$\implies L = e^{-\frac{1}{2}}\left(= \frac{1}{\sqrt{e}}\right)$

Exercise 3

$$\log f(x) = 3x + \log(x^2+1) - \frac{1}{3}\log(x) - \log(\arctan(x))$$
$$\frac{f'(x)}{f(x)} = 3 + \frac{2x}{1+x^2} - \frac{1}{3x} - \frac{1}{\arctan(x)(1+x^2)} = g(x)$$

Hence
$$g(1) = 3 + \frac{2\cdot 1}{1+1^2} - \frac{1}{3\cdot 1} - \frac{1}{\arctan(1)(1+1^2)} = 3 + 1 - \frac{1}{3} - \frac{1}{(\pi/4)(2)} = \frac{11}{3} - \frac{2}{\pi}$$

Exercise 4

$$f'(x) = \frac{\cos(x)}{1+\sin^2(x)}$$

Mean Value Theorem (MVT) requires $\exists c \in (0, \frac{\pi}{2})$ such that
$$f'(c) = \frac{f(\frac{\pi}{2}) - f(0)}{\frac{\pi}{2} - 0}$$
$$\frac{\cos(c)}{1+\sin^2(c)} = \frac{\arctan\left(\sin\frac{\pi}{2}\right) - \arctan(\sin 0)}{\frac{\pi}{2} - 0} = \frac{\arctan(1) - \arctan(0)}{\pi/2} = \frac{\pi/4 - 0}{\pi/2} = \frac{1}{2}$$

which simplifies to
$$2\cos(c) = 1 + \sin^2(c) = 1 + (1 - \cos^2(c)) = 2 - \cos^2(c)$$
$$\cos^2(c) + 2\cos(c) - 2 = 0$$

Hence$\cos(c) = -1 \pm \sqrt{3}$
Since $|\cos c| \leq 1$, we must choose the positive root $\cos c = \sqrt{3} - 1$, $c = \arccos(\sqrt{3} - 1)$

Exercise 5

Apply $\frac{d}{dx}$ to both sides of $y^3 - 2y = 3x^2 - 4x$, we have
$$3y^2\frac{dy}{dx} - 2\frac{dy}{dx} = 6x - 4$$

At $(x, y) = (1, 1)$, we have
$$\frac{dy}{dx} = \frac{6\cdot 1 - 4}{3\cdot 1^2 - 2} = \frac{2}{1} = 2$$

so the function is increasing.
Apply $\frac{d}{dx}$ again to $3y^2\frac{dy}{dx} - 2\frac{dy}{dx} = 6x - 4$:
$$6y\left(\frac{dy}{dx}\right)^2 + 3y^2\frac{d^2y}{dx^2} - 2\frac{d^2y}{dx^2} = 6$$

At $(x, y) = (1, 1)$ and $\frac{dy}{dx} = 2$, we have
$$6\cdot 1\cdot (2)^2 + 3\cdot 1^2\frac{d^2y}{dx^2} - 2\frac{d^2y}{dx^2} = 6$$
$$24 + \frac{d^2y}{dx^2} = 6 \implies \frac{d^2y}{dx^2} = -18$$

Since $\frac{d^2y}{dx^2} = -18 < 0$, the function is concave.

Exercise 6

$g'(5) = (f^{-1}\cdot f^{-1})'(5) = (f^{-1})'(f^{-1}(5)) \cdot (f^{-1})'(5) = (f^{-1})'(1)\cdot(f^{-1})'(5)$
Using the inverse function theorem: $(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$.
$$(f^{-1})'(1)\cdot(f^{-1})'(5) = \frac{1}{f'(f^{-1}(1))}\cdot\frac{1}{f'(f^{-1}(5))}$$

Using the given values:
$$g'(5) = \frac{1}{f'(7)}\cdot\frac{1}{f'(1)} = \frac{1}{-3/2}\cdot\frac{1}{-4/3} = \left(-\frac{2}{3}\right)\cdot\left(-\frac{3}{4}\right) = \frac{6}{12} = \frac{1}{2}$$

Exercise 7

Since g is continuous at $x = 0$, we must have

$$\begin{aligned} a &= \lim_{x\to 0} g(x) \\ &= \lim_{x\to 0}\left(\frac{1}{3xe^{-2x}} - \frac{1}{x(x+3)}\right) \\ &= \lim_{x\to 0} \frac{e^{2x}(x+3)-3}{3x(x+3)} \\ &= \lim_{x\to 0} \frac{e^{2x}(2x+6+1)}{3(2x+3)} \\ &= \frac{7}{9} \end{aligned}$$

Exercise 8

(i)

Let $f'(x) = 0$, so $\frac{x}{4} - 2 = 0$ or $4 - x = 0$. $x = 8$ or $x = 4$.

The critical points are $x = 4$ and $x = 8$.

(ii)

The intervals of increasing/decreasing can be found by checking the sign of $f'(x)$.

$$f'(x) = \left(\frac{x-8}{4}\right)^3 (4-x) = -\frac{1}{64}(x-8)^3(x-4)$$

- $x \in (-\infty, 4)$: $f'(x) = -(-)(-) = -$, so f is decreasing.
- $x \in (4, 8)$: $f'(x) = -(-)(+) = +$, so f is increasing.
- $x \in (8, \infty)$: $f'(x) = -(+)(+) = -$, so f is decreasing.

f is increasing over $(4, 8)$. f is decreasing over $(-\infty, 4), (8, \infty)$.

(iii)

Find $f''(x) = -\frac{1}{16}(x-5)(x-8)^2$, $\quad f''(x) = 0$ at $x = 5$ and $x = 8$.

- $x \in (-\infty, 5)$: $f''(x) = -(-)(+) = +$, so f is convex.
- $x \in (5, 8)$: $f''(x) = -(+)(+) = -$, so f is concave.
- $x \in (8, \infty)$: $f''(x) = -(+)(+) = -$, so f is concave.

f is convex over $(-\infty, 5)$. f is concave over $(5, \infty)$.

(iv)

By the first derivative test, f has a local min at $x = 4$, a local max at $x = 8$.

(v)

Inflection points occur where $f''(x) = 0$ and $f''(x)$ changes sign.
At $x = 5$, $f''(x)$ changes from $+$ to $-$. $x = 5$ is an inflection point.

Exercise 9

$$f'(x) = \frac{(x^2+2x+3)\frac{2}{3}x^{-1/3} - x^{2/3}(2x+2)}{(x^2+2x+3)^2} = \frac{-2(2x+3)(x-1)}{3x^{1/3}(x^2+2x+3)^2}$$

Critical points are where $f'(x) = 0$ or $f'(x)$ is undefined:

- $f'(x) = 0$: $(2x+3)(x-1) = 0 \implies x = -3/2$ or $x = 1$.
- $f'(x)$ is undefined: $x = 0$.

Candidate points: endpoints $x = -2, x = 2$, and critical points $x = -3/2, x = 0, x = 1$.
Evaluation of $f(x)$ at candidate points:

- $f(-2) = \frac{\sqrt[3]{(-2)^2}}{(-2)^2+2(-2)+3} = \frac{\sqrt[3]{4}}{4-4+3} = \frac{\sqrt[3]{4}}{3}$
- $f(2) = \frac{\sqrt[3]{2^2}}{2^2+2(2)+3} = \frac{\sqrt[3]{4}}{4+4+3} = \frac{\sqrt[3]{4}}{11}$
- $f(0) = \frac{\sqrt[3]{0^2}}{0+0+3} = 0$ global min
- $f(1) = \frac{\sqrt[3]{1^2}}{1+2+3} = \frac{1}{6}$
- $f(-3/2) = (\frac{2}{3})^{\frac{4}{3}}$ global max